

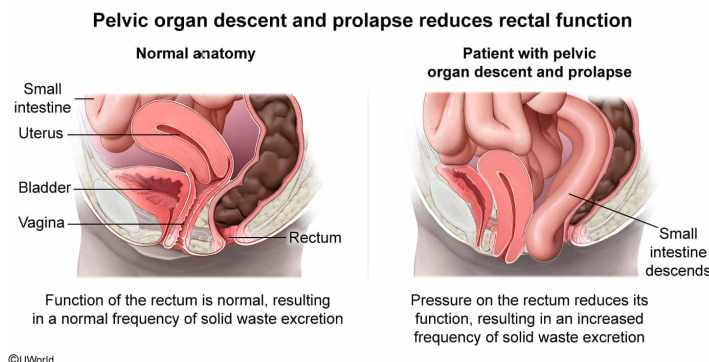
Pelvic organ descent and prolapse is a pathology in which abdominopelvic organs move downward and put pressure on lower organs. Pelvic organ descent and prolapse would most likely cause the function of the rectum to be:

- ☐ A. enhanced, resulting in a decreased frequency of solid waste excretion [6%]
- ✓ ☒ B. reduced, resulting in an increased frequency of solid waste excretion [46%]
- ☐ C. enhanced, resulting in an increased quantity of water reabsorbed from the feces [15%]
- ☐ D. reduced, resulting in a decreased quantity of water reabsorbed from the feces [31%]

Correct

45% answered correctly

Explanation:



Ingested food travels through the **gastrointestinal (GI) tract**, composed of the esophagus, **stomach**, **small intestine**, and **large intestine**. **Chemical** and mechanical digestion occur in the earlier sections of the GI tract; therefore, once ingested products reach the large intestine, all digestion is complete.

The large intestine is a tube surrounded by **smooth muscle** through which undigestible material travels via **peristalsis**. **Three sections** compose the large intestine: the cecum, the colon, and the rectum. The colon is responsible for reabsorption of electrolytes (eg, sodium,

chloride) and water from the undigestible material. As this occurs, the material is compacted into solid waste (feces), which is stored in the last section of the large intestine (the rectum) prior to elimination from the body.

In this question, pelvic organ descent and prolapse is a pathology in which the abdominopelvic organs (eg, small intestine, [uterus](#), [bladder](#)) move downward and put pressure on the lower organs (eg, rectum).

The rectum is primarily responsible for storage of feces until its excretion. If pressure is put on the rectum, this storage area will be *reduced*. Therefore, a patient with pelvic organ descent and prolapse will likely experience **reduced rectal function**, resulting in an **increased frequency of solid waste excretion**.

(Choices A and C) Pressure on the rectum would result in reduced, *not* enhanced, rectal function.

(Choice D) The colon, *not* the rectum, is involved in reabsorbing water from the feces.

Educational objective:

The large intestine includes the cecum, the colon, and the rectum. The rectum is responsible for storage of solid waste (feces) before it is eliminated from the body.

Time spent:0 sec

Subject:

Biology

QID:403383

System:

Digestion and Excretion